AP CALCULUS AB

Unit 1 • Lesson 1.4 practice

Lesson 1.4 • Estimating Limit Values from Tables

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

x	1.9	1.99	1.999	2.001	2.01	2.1
$f(x)$	0.47	0.497	0.4997	0.5003	0.503	0.53

Which is the best estimate of $\lim_{x \to 2} f(x)$?

(A) 0.5

(B) 0.4997

(C) 2

(D) It does not exist.

Question 2

NO CALCULATOR

As $x \to 3^-$, the values of $g(x)$ are 1.9, 1.99, 1.999; as $x \to 3^+$, they are -0.9, -0.99, -0.999. What is $\lim_{x \to 3} g(x)$?

(A) 2

(B) -1

(C) 1

(D) It does not exist.

Question 3

NO CALCULATOR

As $x \to 0$ from both sides, $h(x)$ takes the values 10, 100, 1000, 10000. Which statement is true?

(A) $\lim_{x \to 0} h(x) = 10000$

(B) h increases without bound near 0; the limit does not exist as a real number.

(C) $\lim_{x \to 0} h(x) = 0$

(D) $h(0) = 10000$

Question 4

NO CALCULATOR

Which is true about using a table to find a limit?

(A) A table always gives the exact limit.

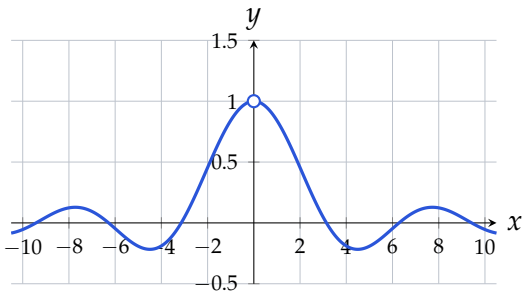
(B) A table cannot suggest a limit.

(C) A table suggests a value but does not prove the limit.

(D) A table needs only values on one side.

Question 5 (graph) NO CALCULATOR

The graph of $y = \dfrac{\sin x}{x}$ is shown; it has a hole at $x = 0$. Which is the best estimate of $\lim\limits_{x \to 0} \dfrac{\sin x}{x}$?



(A) 0

(B) π

(C) 1

(D) It does not exist.

Question 6 (Free response) NO CALCULATOR

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x) = \frac{e^x - 1}{x}$	0.9516	0.9950	0.9995	1.0005	1.0050	1.0517

(a) Use the table to estimate $\lim\limits_{x \to 0} f(x)$.

(b) Is $f(0)$ defined? Does this affect your answer to (a)?

(c) A function g has $g(1.9) = 1.9$, $g(1.99) = 1.99$, $g(1.999) = 1.999$, $g(2.1) = 3.1$, $g(2.01) = 3.01$, $g(2.001) = 3.001$. Estimate the one-sided limits of g at $x = 2$. Does $\lim\limits_{x \to 2} g(x)$ appear to exist?